TEAM NISHKARSH

SMART KOPARGOAN HACKATHON

CleanKop AI

AI, IoT & GPS-Based Smart Garbage Collection and Monitoring System

PROBLEM STATEMENT ID	PS CATEGORY
SKH21	**Software**
PROBLEM STATEMENT TITLE	TEAM ID
GPS-Based Smart Garbage Collection and Monitoring System	**T4M4VZ**
THEME	TEAM NAME
Open Innovation	**Team Nishkarsh**

Team Members: Parth Desai (Leader) • Divya Pawar • Shreyash Patil • Tanavi Pawar

Team Nishkarsh

IDEA TITLE

CleanKop AI — Proposed Solution

A unified IoT + AI platform that gives every bin a digital identity, every collection route a data-driven plan, and every citizen complaint a transparent trail — replacing today's manual, fixed-schedule waste collection.

PROBLEMS WE SOLVE

- No real-time fill data — collections run on fixed schedules, not actual need
- No centralized tracking — zero accountability or data synchronization
- No digital complaints — issues stay informal, untracked, unprioritized
- No feedback loop — performance is never measured or improved

HOW CLEANKOP AI ADDRESSES IT

- Ultrasonic + ESP32 sensors report live bin fill % over Wi-Fi/GSM
- TSP nearest-neighbor + 2-opt engine plans the shortest priority route
- Citizen app + ticketed complaints with Submitted → Resolved workflow
- Live admin dashboard tracks fuel saved, resolution time & performance

Team Nishkarsh

TECHNICAL APPROACH

Technology Stack & Methodology

Frontend
React.js + Tailwind CSS

Backend
Node.js + Express

Database
MongoDB / PostgreSQL + PostGIS

Maps
Leaflet.js + OpenStreetMap

Real-time & Alerts
Socket.io, Nodemailer / Twilio

Hosting
Vercel + Render + Mapbox APIs

IMPLEMENTATION METHODOLOGY

1

Sensor → Cloud

ESP32 + ultrasonic sensor sends fill-level data to the backend every few minutes (hackathon: simulator script)

2

Store & Alert

Backend stores bin data in the database; bins over 80% trigger an automatic alert

3

Optimize Route

TSP nearest-neighbor + 2-opt algorithm ranks red/yellow/green bins and outputs the shortest route

4

Collect & Update

Driver app tracks GPS live and marks bins collected; status syncs to admin in real time

5

Resolve Complaints

Citizen reports flow through Submitted → Assigned → In Progress → Resolved, with 48-hr escalation

Team Nishkarsh

TECHNICAL APPROACH

System Architecture

HOW TO READ IT

IoT layer

Bin sensors + ESP32 push fill data to the backend

Backend core

Node.js/Express handles auth, routing, complaints, alerts

Data layer

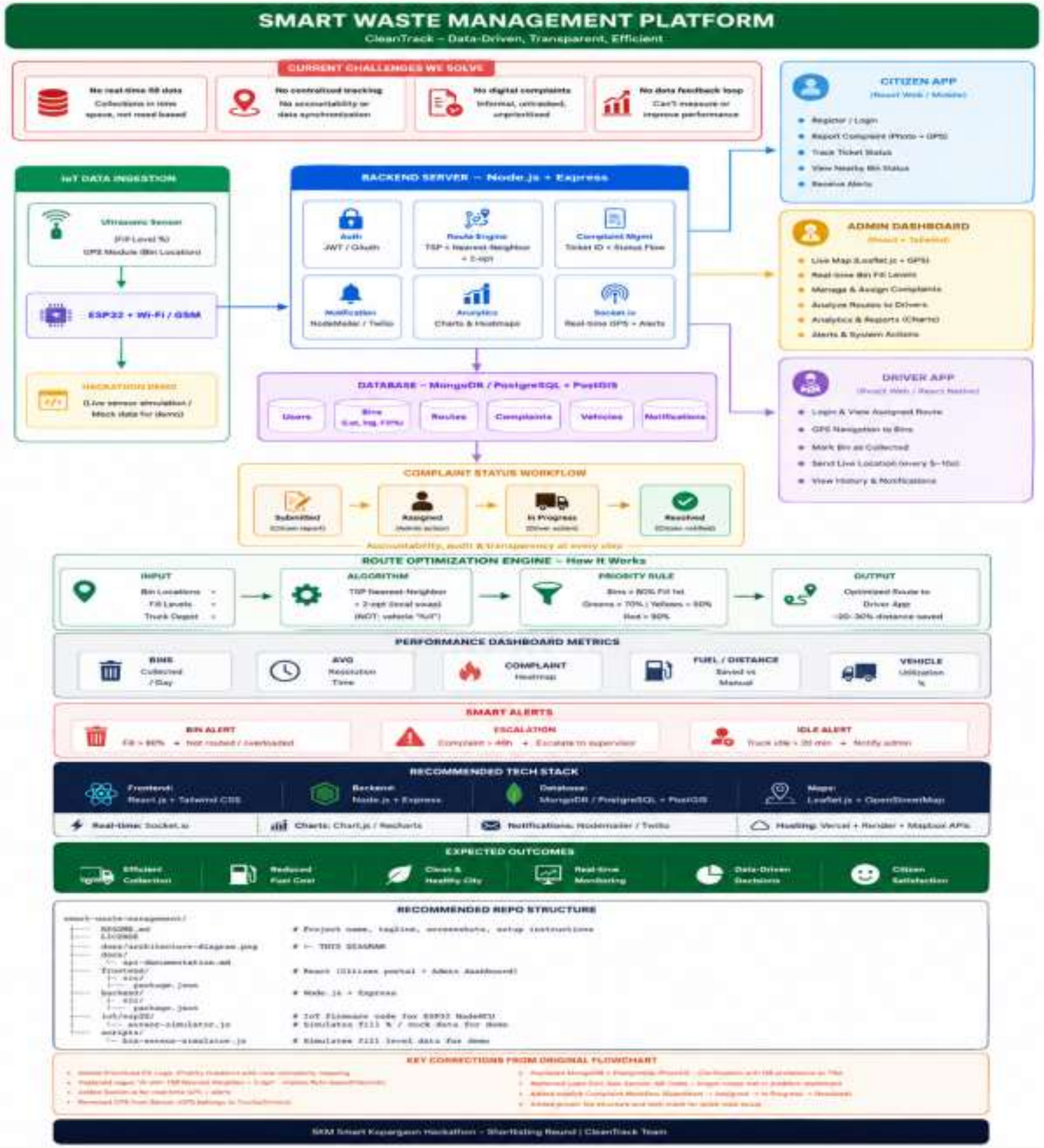
MongoDB/PostgreSQL + PostGIS stores bins, routes, users

Three apps

Citizen, Admin & Driver apps share one live backend

Route engine

TSP + 2-opt turns priority bins into an optimized path



Team Nishkarsh

FEASIBILITY AND VIABILITY

Can We Build & Run This?

FEASIBILITY

- ESP32 + ultrasonic sensors are low-cost, off-the-shelf, and hackathon-ready
- Full stack (React, Node.js, MongoDB/PostgreSQL) is mature, documented, and free to prototype on
- TSP nearest-neighbor + 2-opt is lightweight enough to run on a standard server in real time
- Firmware-ready architecture: today's simulator script swaps for real sensors with no redesign

RISKS → MITIGATION STRATEGY

Per-bin hardware cost

Phase rollout — pilot in a few high-traffic wards before city-wide scale-up

Patchy rural connectivity

GSM fallback alongside Wi-Fi so bins report even without broadband

Driver / citizen adoption

Simple mobile-first apps + training and incentive program for drivers

Sensor data accuracy

Threshold smoothing + manual override option in the admin dashboard

Team Nishkarsh

IMPACT AND BENEFITS

Who Gains, and How

Target audience: Kopargaon municipal corporation, sanitation drivers, and citizens across the city.

Efficient Collection

Priority routing (red → yellow → green bins) cuts wasted trips

Reduced Fuel Cost

20–30% distance saved vs. fixed manual routes

Clean & Healthy City

Bins emptied before overflow — fewer health & odor complaints

Real-time Monitoring

Admins see live fill levels, trucks, and complaints on one map

Data-Driven Decisions

Analytics and heatmaps guide staffing and future bin placement

Citizen Satisfaction

Transparent ticket status builds trust in the system

Team Nishkarsh

RESEARCH AND REFERENCES

Grounded in Established Tools & Methods

ESP32 + Ultrasonic Sensing

Espressif ESP32 technical reference & HC-SR04 ultrasonic sensor documentation for bin fill-level sensing

Route Optimization

Traveling Salesman Problem — Nearest-Neighbor construction heuristic refined with 2-opt local search

Geospatial Data

PostgreSQL + PostGIS documentation for storing and querying bin/route geo-coordinates

Mapping & Live Tracking

Leaflet.js and OpenStreetMap documentation for the live map and route rendering

Real-time Communication

Socket.io documentation for live GPS updates, bin alerts, and dashboard sync

Reference Municipal Models

Smart-city solid-waste-management case studies (IoT bin sensors + optimized collection routing)

RESEARCH AND REFRENCES

DETAILS

ESP32 & Ultrasonic Sensor

Espressif Systems. *ESP-IDF Programming Guide (Official Documentation).*

HC-SR04 Ultrasonic Sensor Datasheet (Components101 / Manufacturer Documentation).

Route Optimization

Traveling Salesman Problem (TSP): Nearest Neighbor + 2-Opt Heuristic.

MDPI, *Applied Sciences* (2023).

Geospatial Database

PostgreSQL + **PostGIS Official Documentation**.

Live Mapping

Leaflet.js Official Documentation.

OpenStreetMap Documentation.

Real-Time Communication

Socket.IO v4 Official Documentation

.

Real-World Smart Waste Management Case Studies

Mumbai (Malad), India – IoT Smart Bin Deployment (IIETA Journal): ~20% reduction in garbage truck usage

Team Members: Parth Desai (Leader) • Divya Pawar • Shreyash Patil • Tanavi Pawar